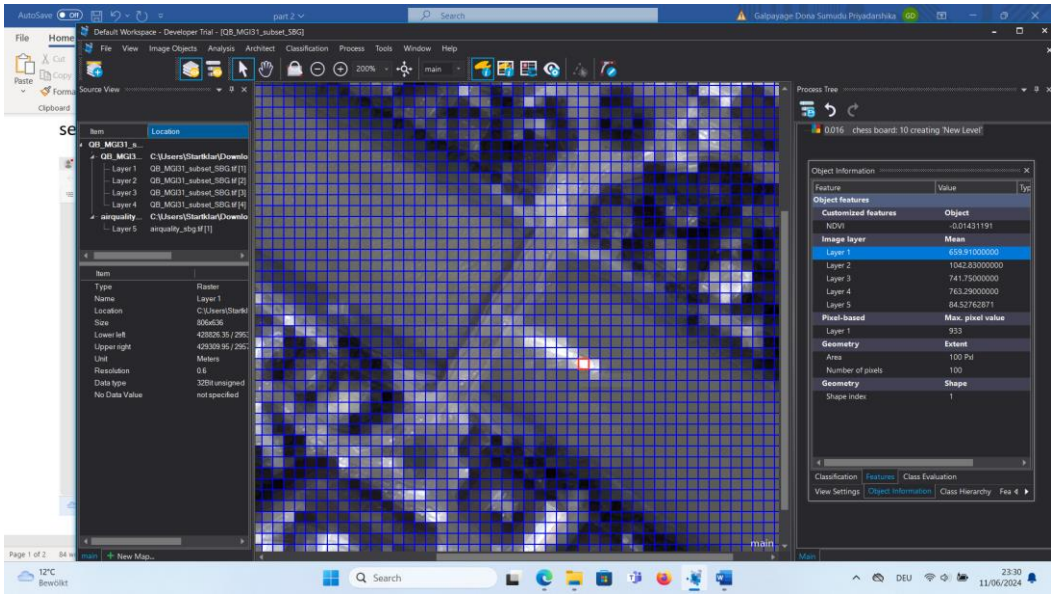
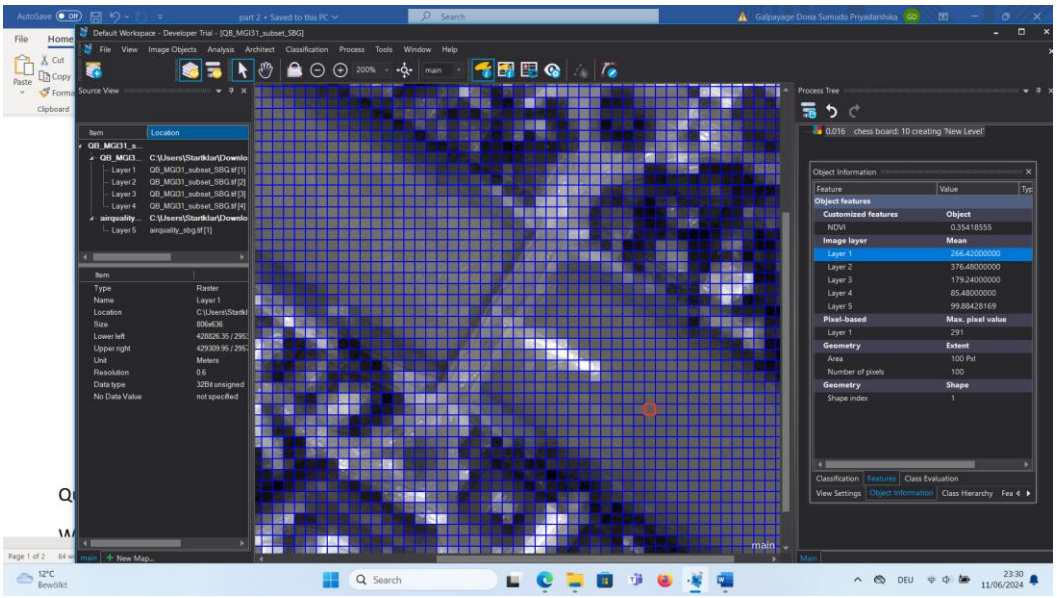
1. **What becomes obvious if you compare the values for the two objects of the chessboard segmentation?**





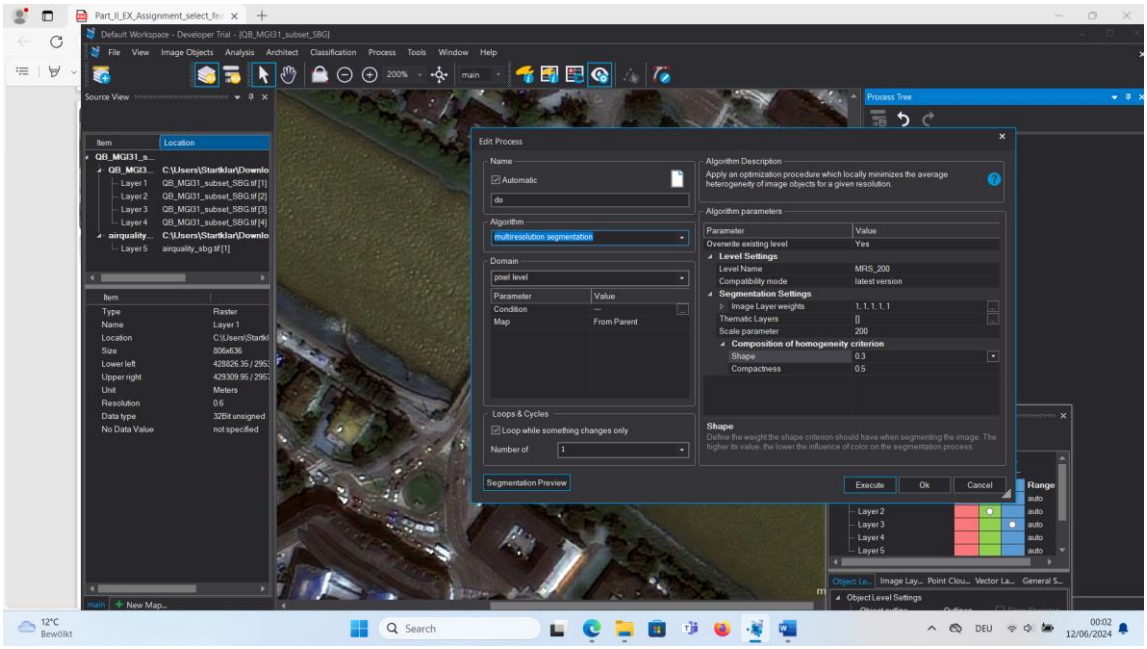
It is obvious that the mean value, maximum pixel value and the NDVI value show differences between the two object features. And the shape index, the area and the pixel values are similar in both the objects.

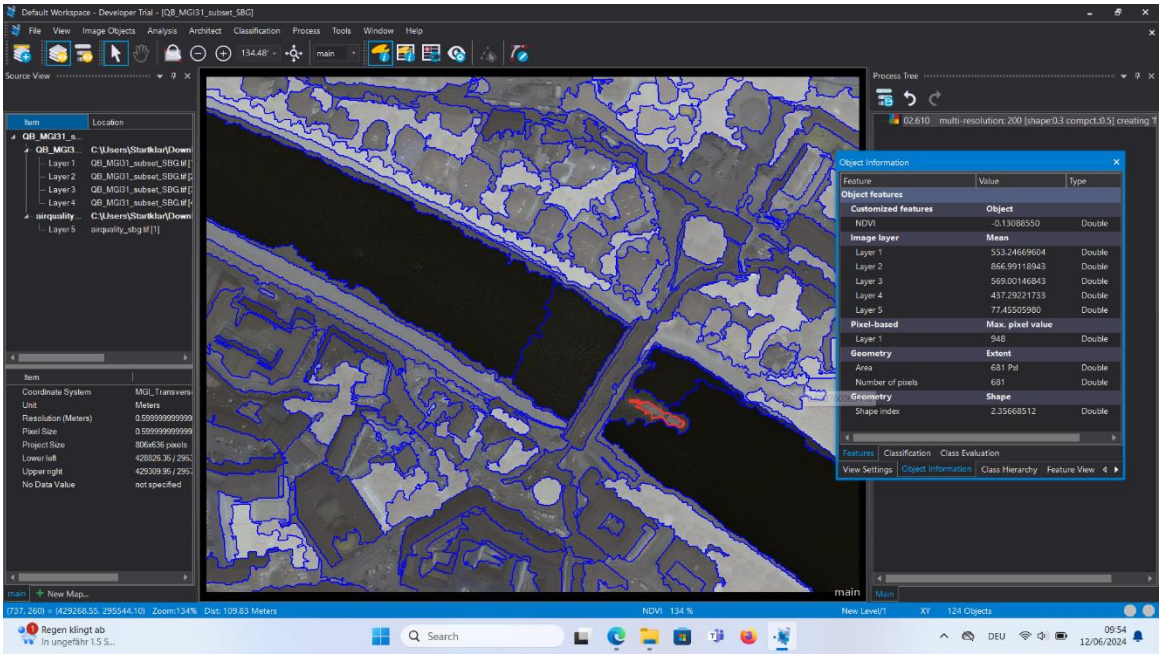
2: Which features don‘t make that much sense in this case?

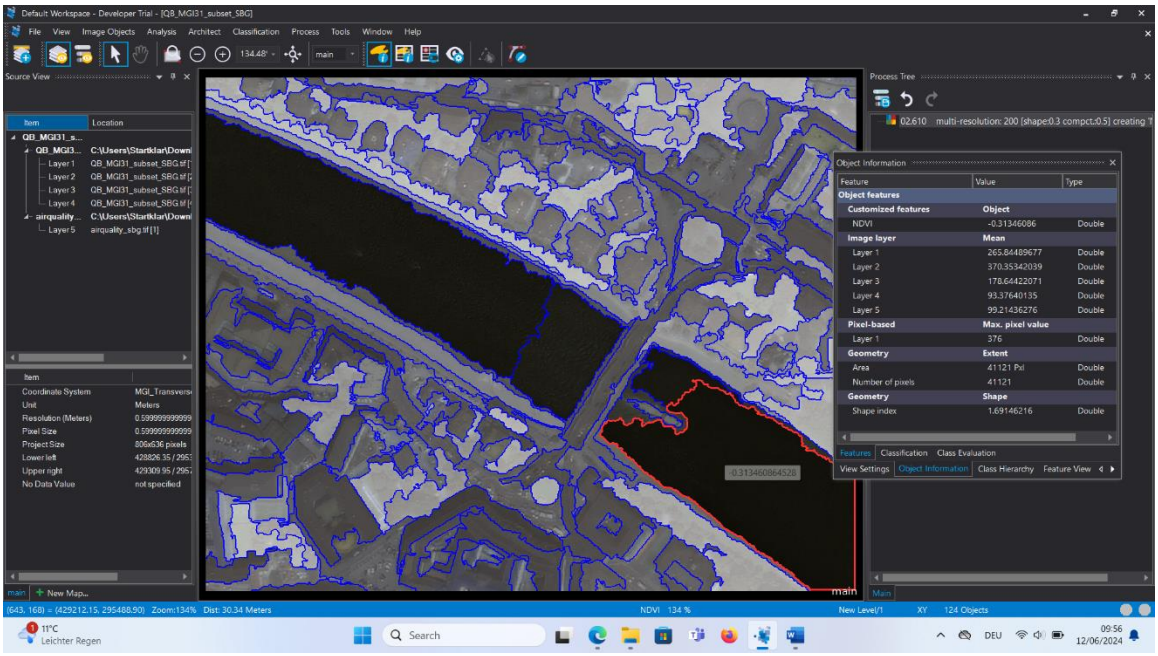
The area [100 pixel], number of pixels[100] and the shape index [1]of the objects do not make that much sense in this case as they do not reflect differences.

Multi resolution segmentation

Steps followed:



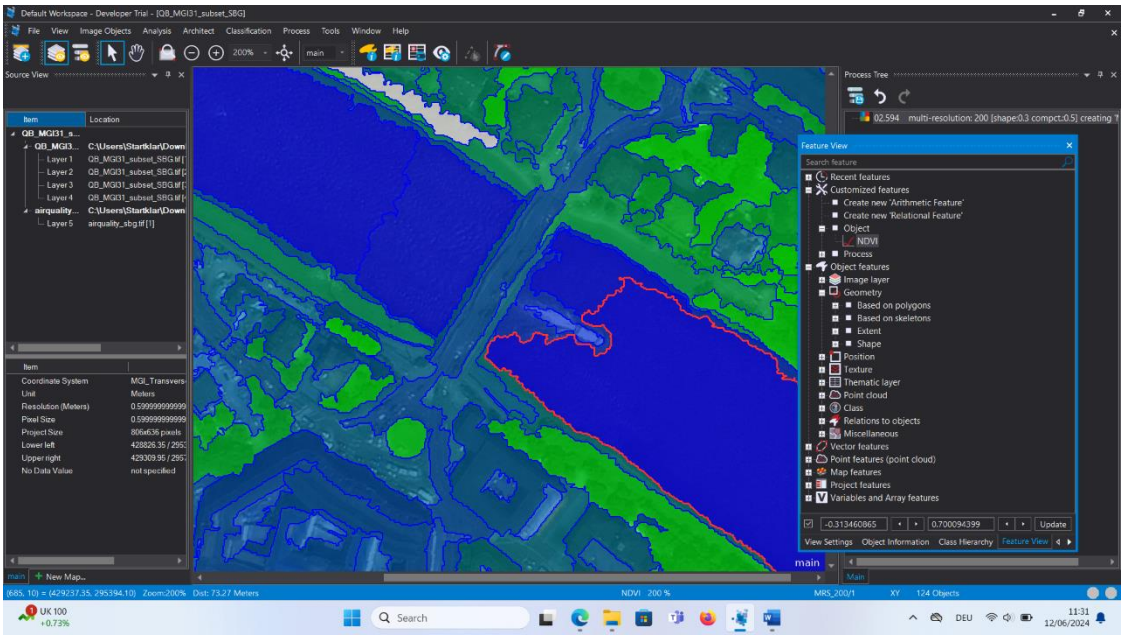


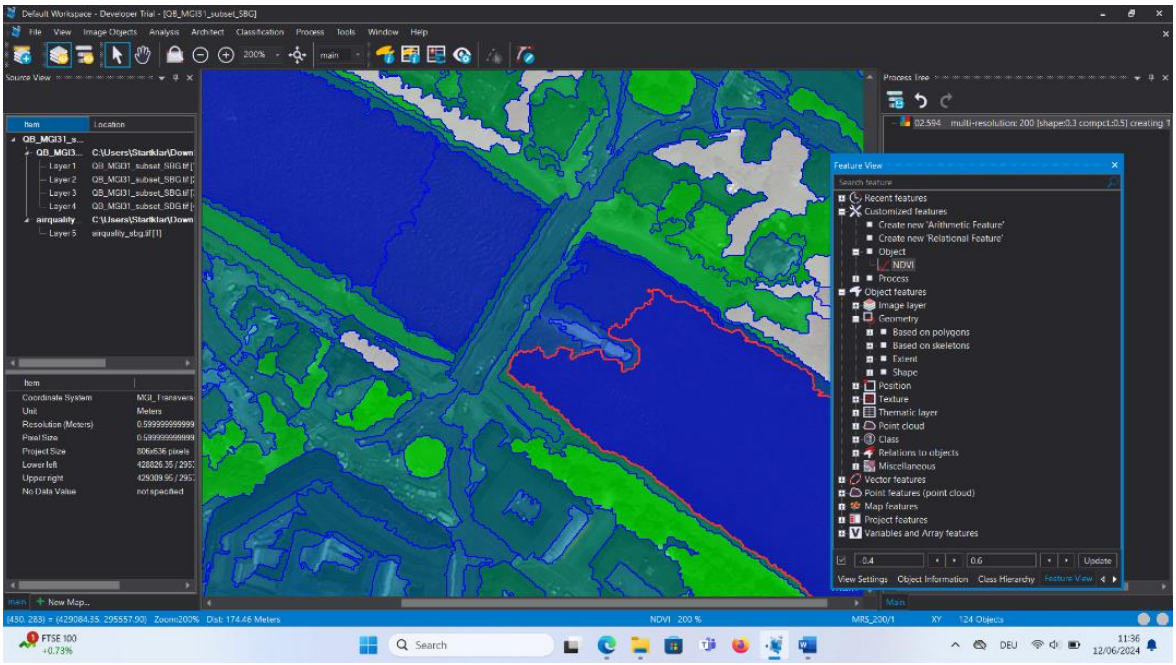


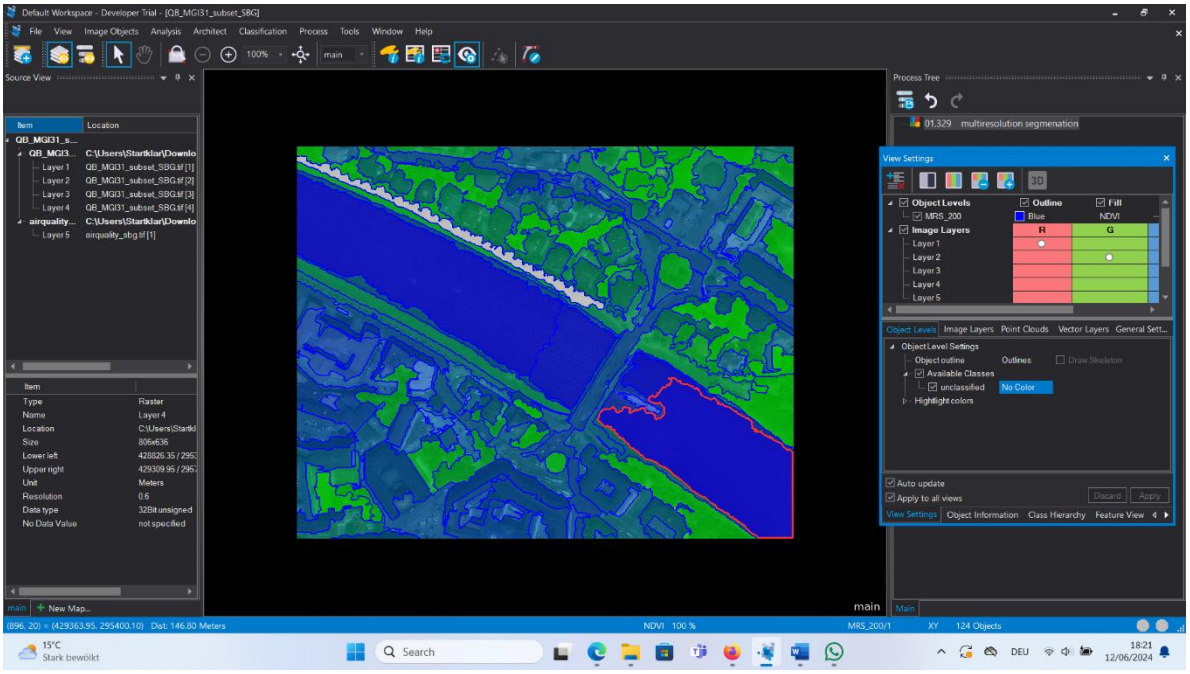
In multi resolution segmentation, mean values, number of pixels, area, shape index and the NDVI values are different for the selected features.

Therefore, it is suggested that the multiresolution segmentation is much better than the chess board segmentation process. Because multi resolution segmentation distinguishes different features from each other based on their specific characteristics.

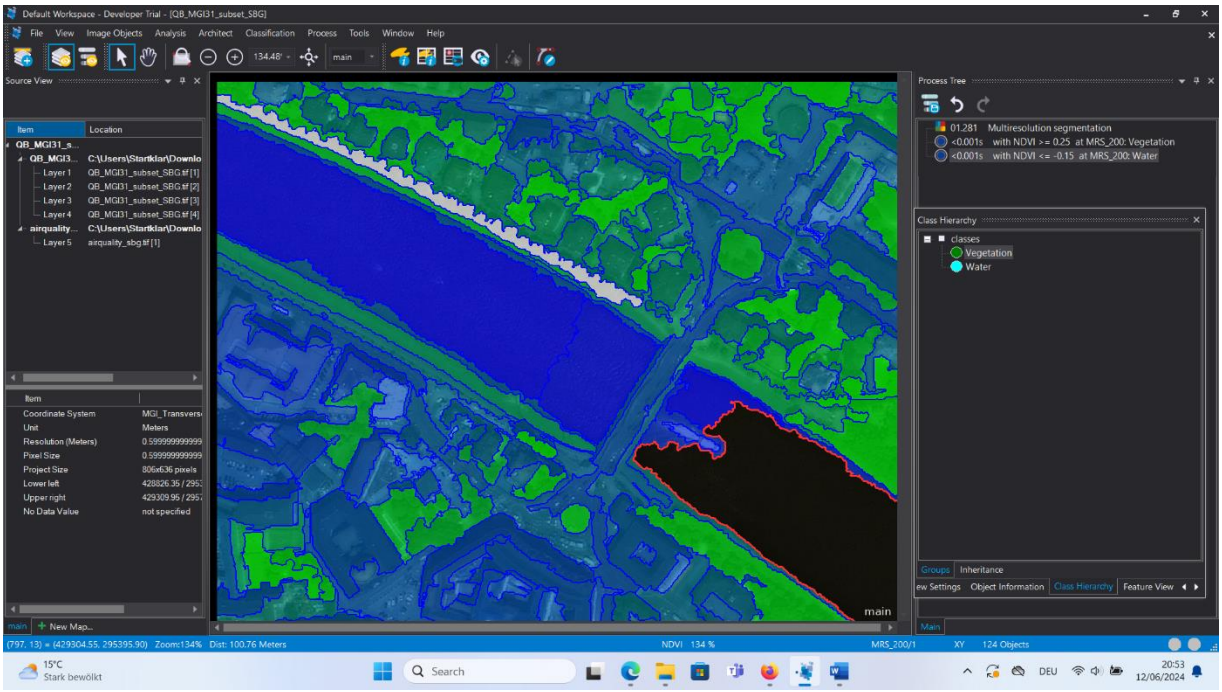
Creating and practicing with NDVI :

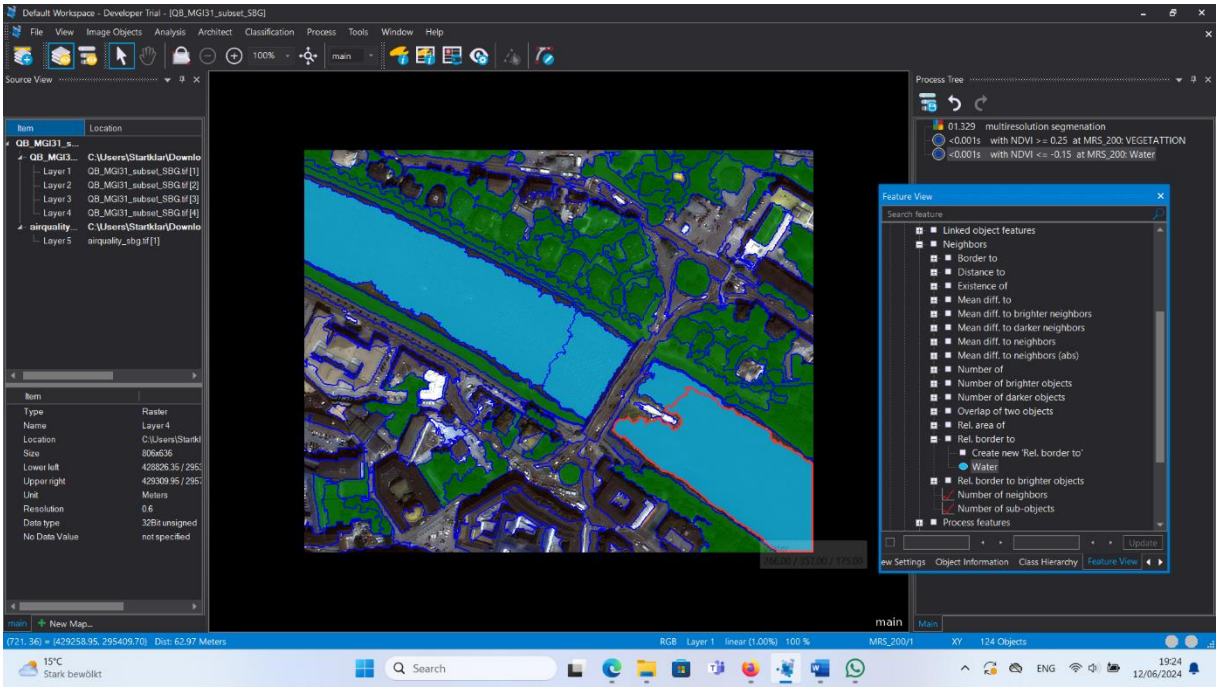


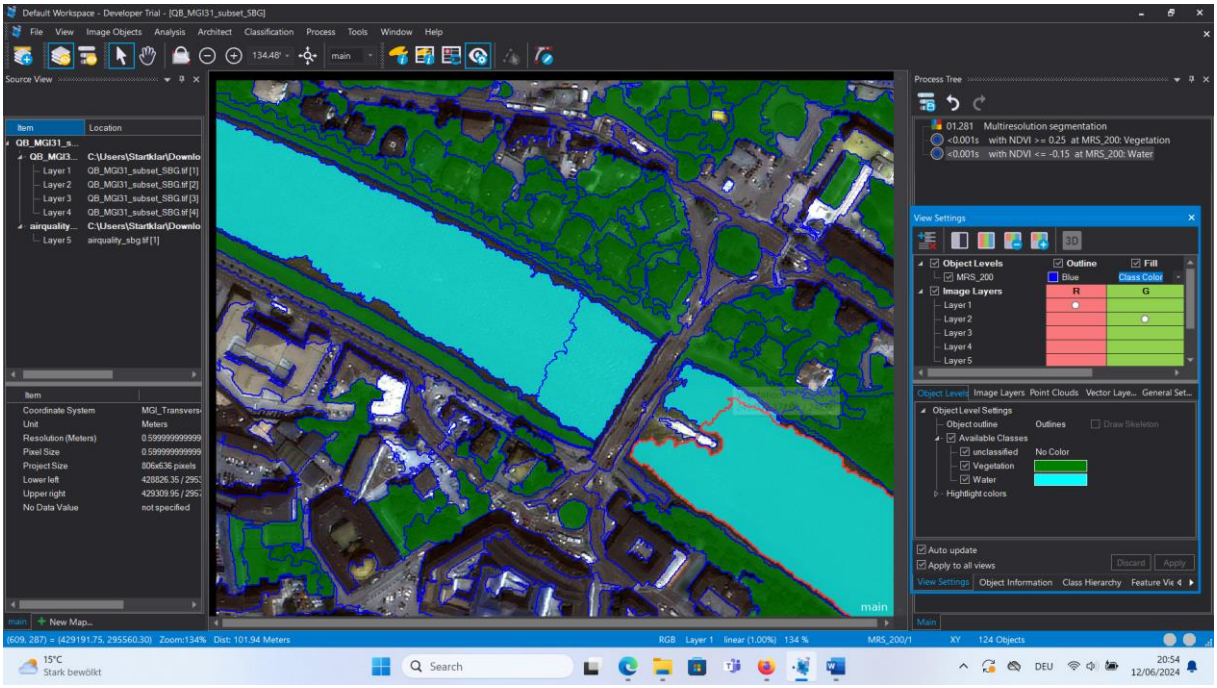

Default Workspace - Developer Trial - [QB_MGI31_subset_SBG]
Source View
Process Tree
Feature View
Recent features
Customized features
Object features
Image layer
Geometry
Vector features
Point features (point cloud)
Map features
Project features
Variables and Array features
Search
11:36
12/06/2024

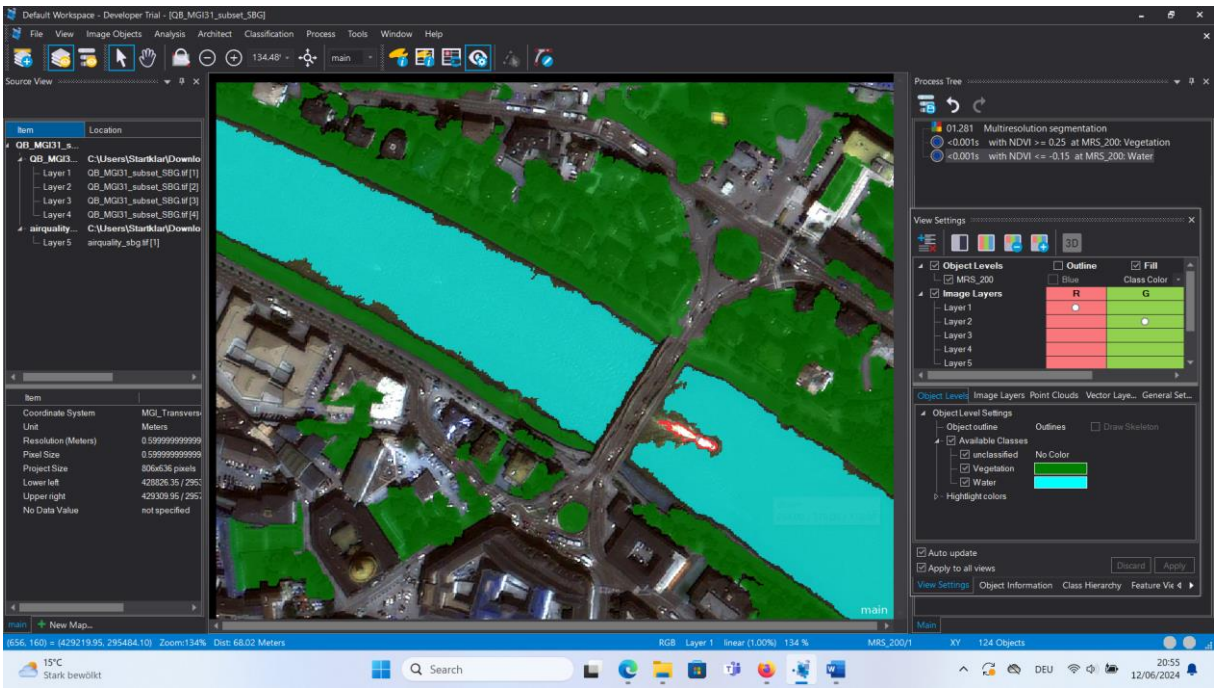

Default Workspace - Developer Trial - [QB_MGI31_subset_SBG]
Source View
Process Tree
01.329 multiresolution segmenation
View Settings
Object Levels
Image Layers
Point Clouds
Vector Layers
General Sett...
main
Search
18:21
12/06/2024

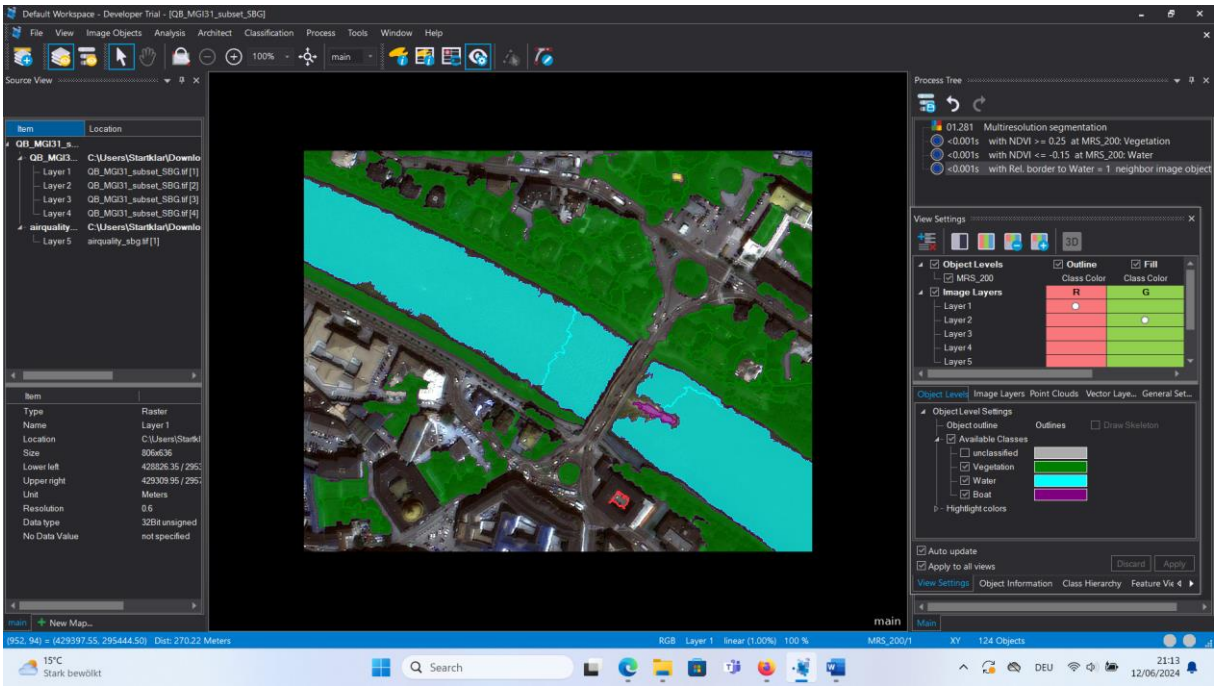
Creating classes

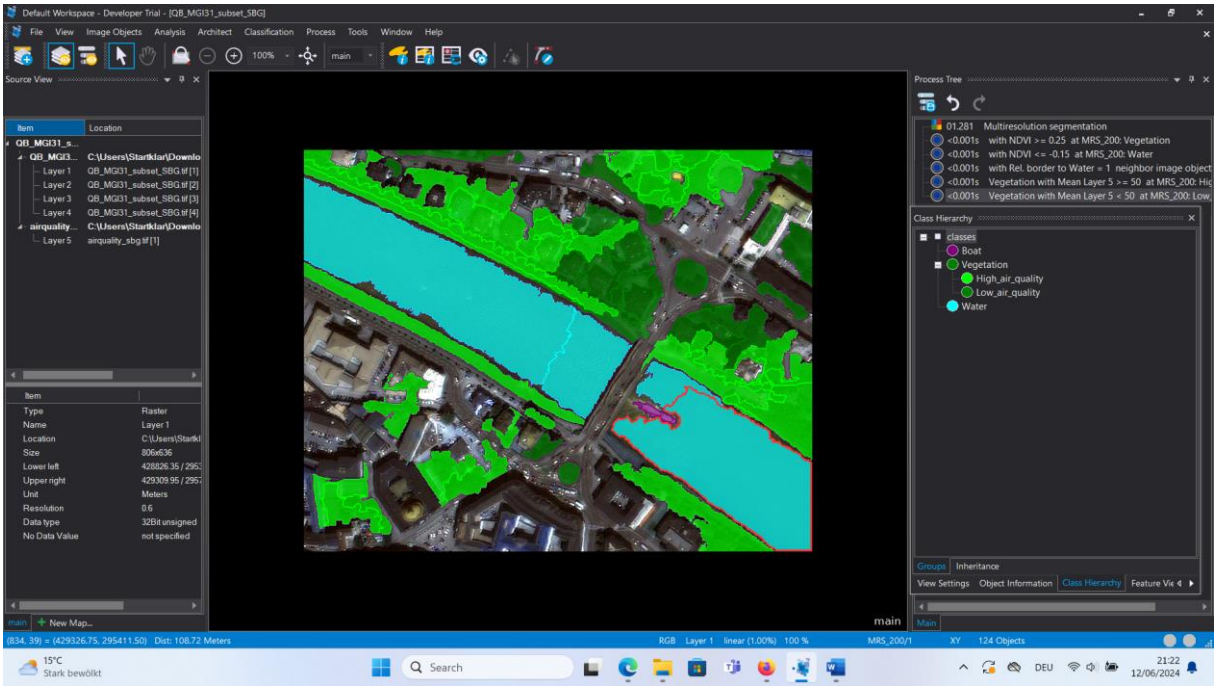





Default Workspace - Developer Trial - [QB_MGI31_subset_SBG]
File View Image Objects Analysis Architect Classification Process Tools Window Help
Source View
Process Tree
01.281 Multiresolution segmentation
<0.001s with NDVI >= 0.25 at MRS_200: Vegetation
<0.001s with NDVI <= -0.15 at MRS_200: Water
View Settings
Object Levels
Image Layers
Object Level Settings
Available Classes
unclassified
Vegetation
Water
Highlight colors
Auto update
Apply to all views
Discard
Apply
main
(609, 287) = (429191.75, 295560.30) Zoom:134% Dist: 101.94 Meters
124 Objects
15°C
Stark bewölkt
Search
DEU
20:54
12/06/2024

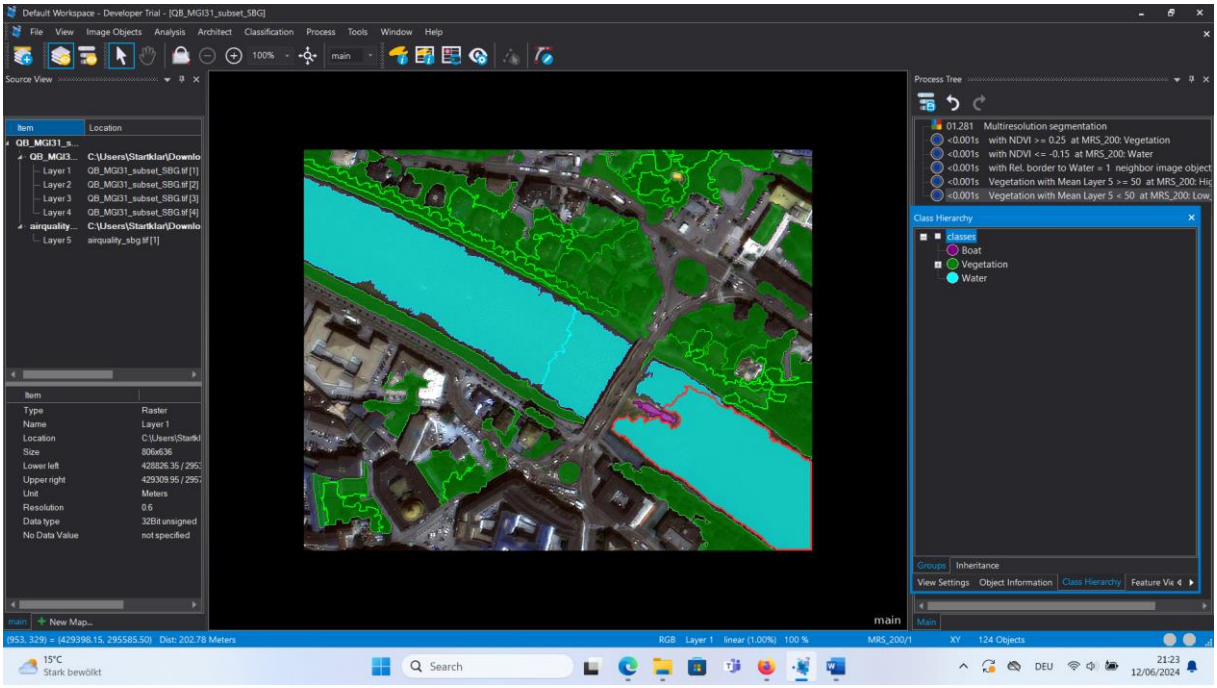

Default Workspace - Developer Trial - [QB_MGI31_subset_SBG]
File View Image Objects Analysis Architect Classification Process Tools Window Help
Source View
Process Tree
01.281 Multiresolution segmentation
<0.001s with NDVI >= 0.25 at MRS_200: Vegetation
<0.001s with NDVI <= -0.15 at MRS_200: Water
View Settings
Object Levels
Image Layers
Object Level Settings
Available Classes
unclassified
Vegetation
Water
Highlight colors
Auto update
Apply to all views
Discard
Apply
main
(656, 160) = (429219.95, 295484.10) Zoom:134% Dist: 68.02 Meters
124 Objects
15°C
Stark bewölkt
Search
DEU
20:55
12/06/2024


Default Workspace - Developer Trial - [QB_MGI31_subset_SBG]
File View Image Objects Analysis Architect Classification Process Tools Window Help
Source View
Process Tree
01.281 Multiresolution segmentation
<0.001s with NDVI >= 0.25 at MRS_200: Vegetation
<0.001s with NDVI <= -0.15 at MRS_200: Water
<0.001s with Rel. border to Water = 1 neighbor image object
View Settings
Object Levels
MRS_200
Image Layers
Layer 1
Layer 2
Layer 3
Layer 4
Layer 5
Object Level Settings
Object outline
Available Classes
unclassified
Vegetation
Water
Boat
Auto update
Apply to all views
main
Search
15°C
Stark bewölkt
DEU
21:13
12/06/2024

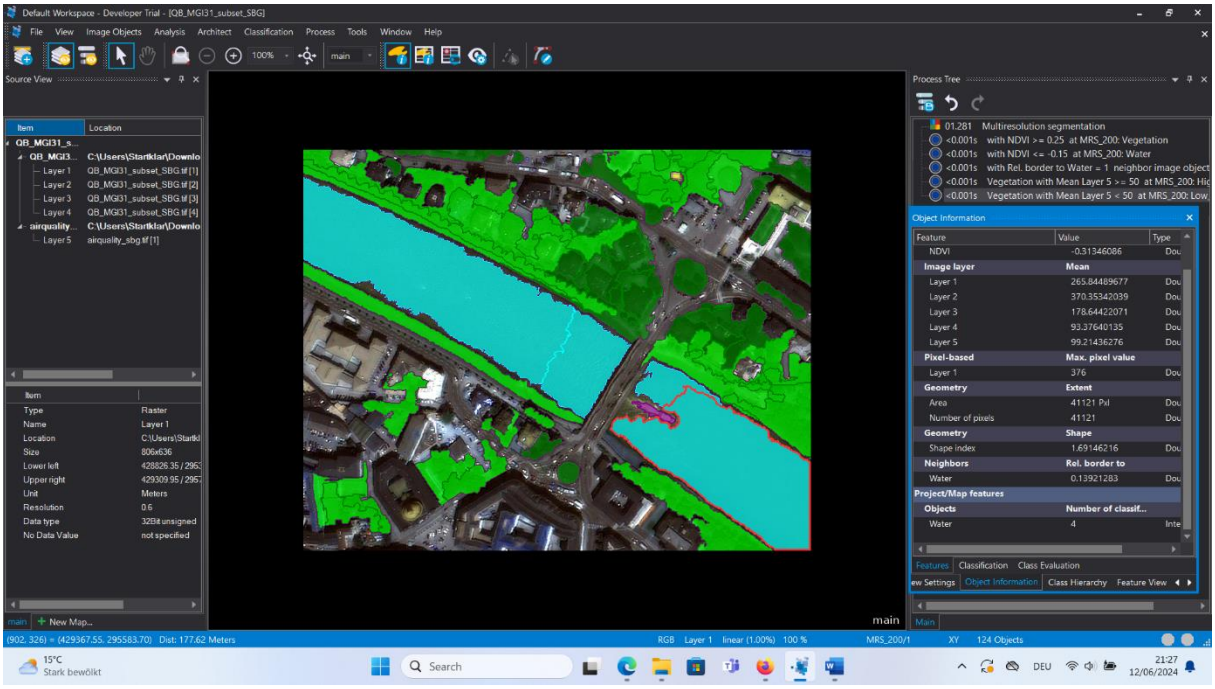

Default Workspace - Developer Trial - [QB_MGI31_subset_SBG]
File View Image Objects Analysis Architect Classification Process Tools Window Help
Source View
Process Tree
01.281 Multiresolution segmentation
<0.001s with NDVI >= 0.25 at MRS_200: Vegetation
<0.001s with NDVI <= -0.15 at MRS_200: Water
<0.001s with Rel. border to Water = 1 neighbor image object
<0.001s Vegetation with Mean Layer 5 >= 50 at MRS_200: Hig
<0.001s Vegetation with Mean Layer 5 < 50 at MRS_200: Low
Class Hierarchy
classes
Boat
Vegetation
High_air_quality
Low_air_quality
Water
main
Search
15°C
Stark bewölkt
DEU
21:22
12/06/2024

What happens if you collapse the vegetation super-class?

If the vegetation superclass collapses the classification disappears as appeared in the following figure.

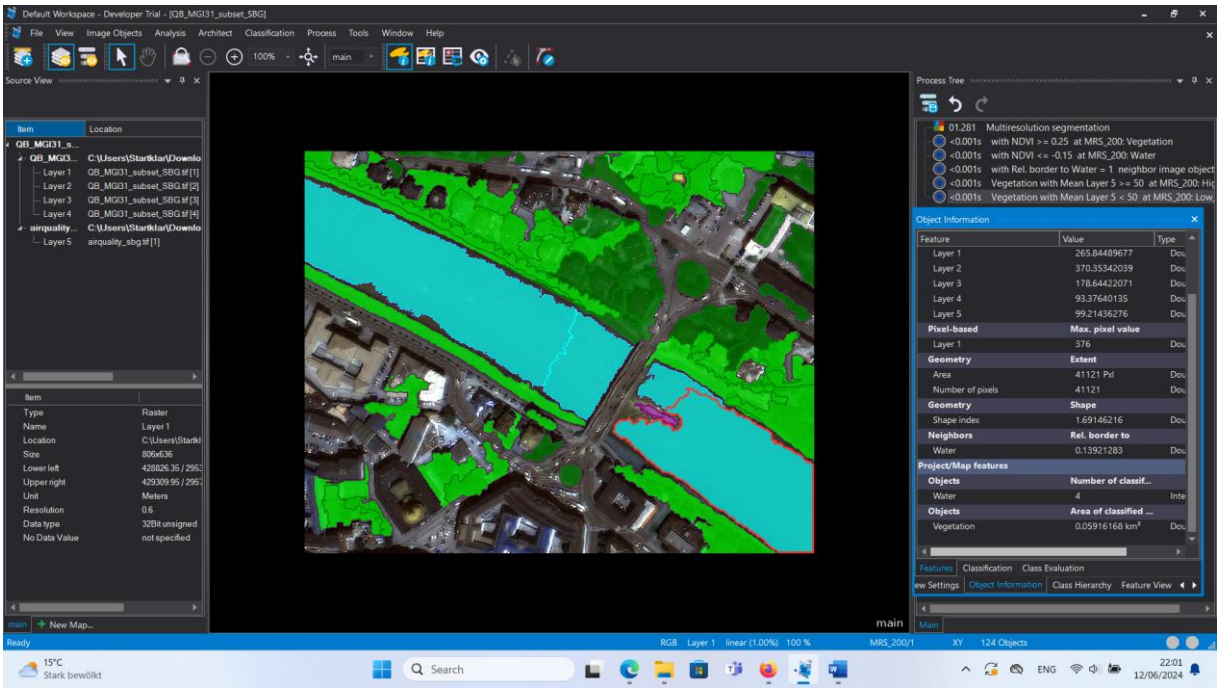


3.How many objects were classified as “water"? There are 4 objects classified as water. See the following figure.

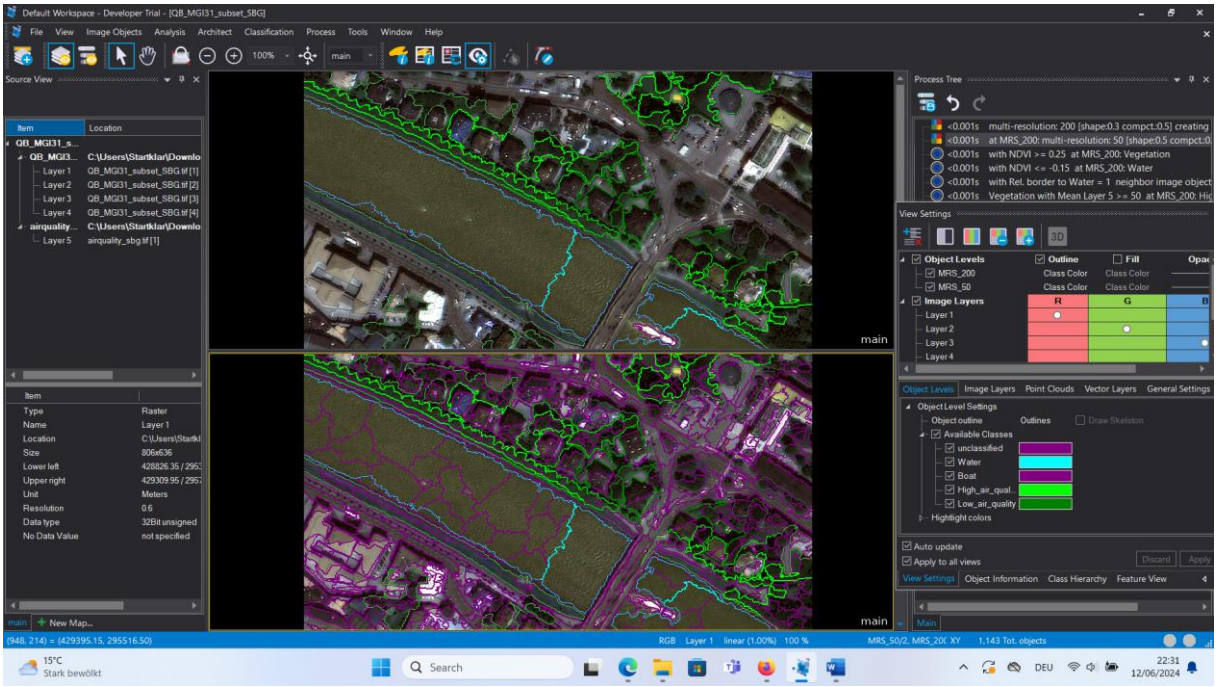


4: **What is the area of the whole vegetation class**

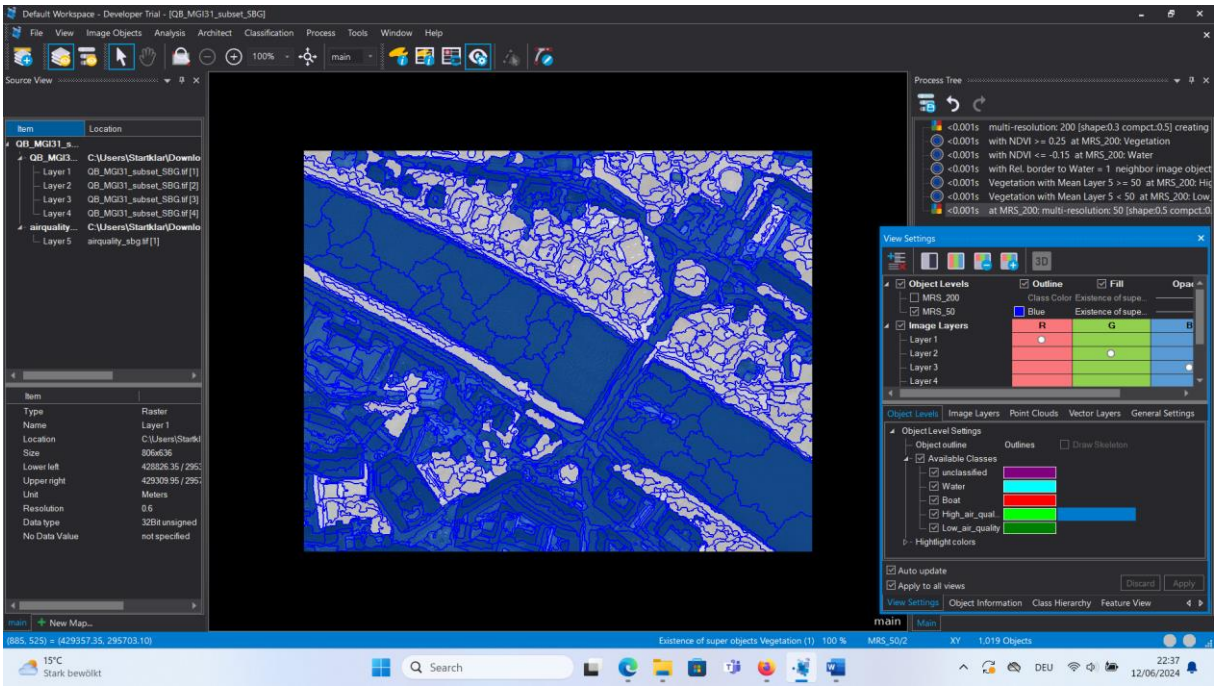
The area of the whole vegetation class is 0.0591 km^2 as in the following figure.

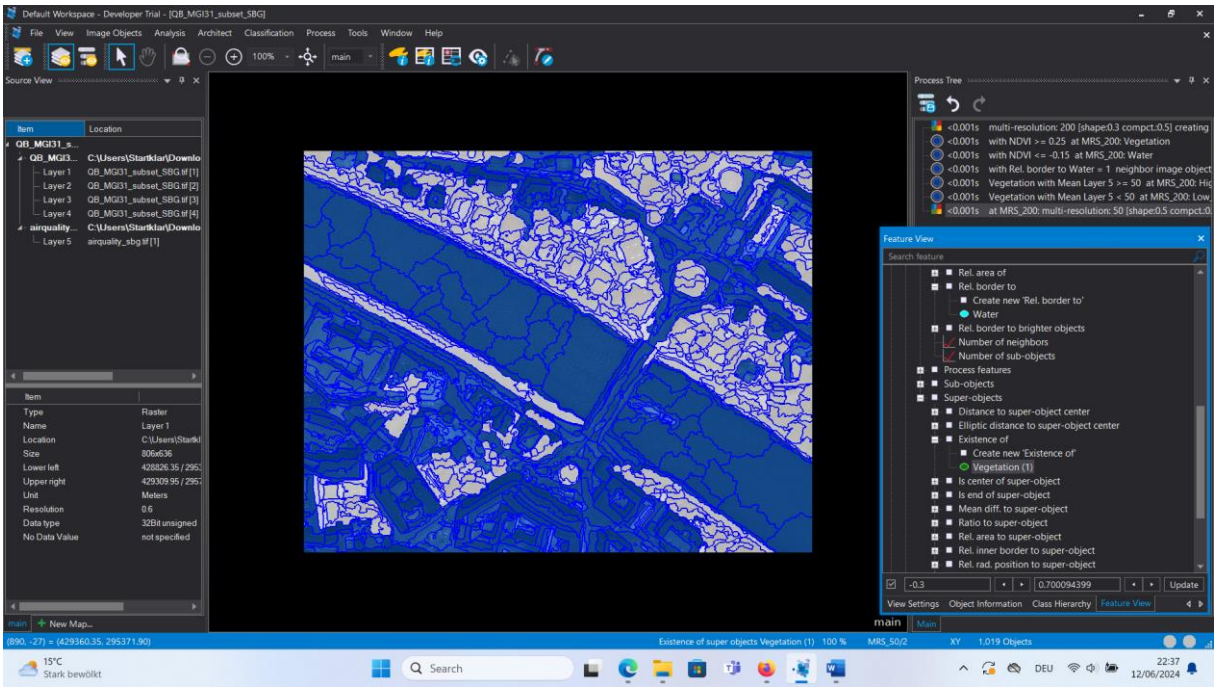


Creating a new level below



Creating relative borders:



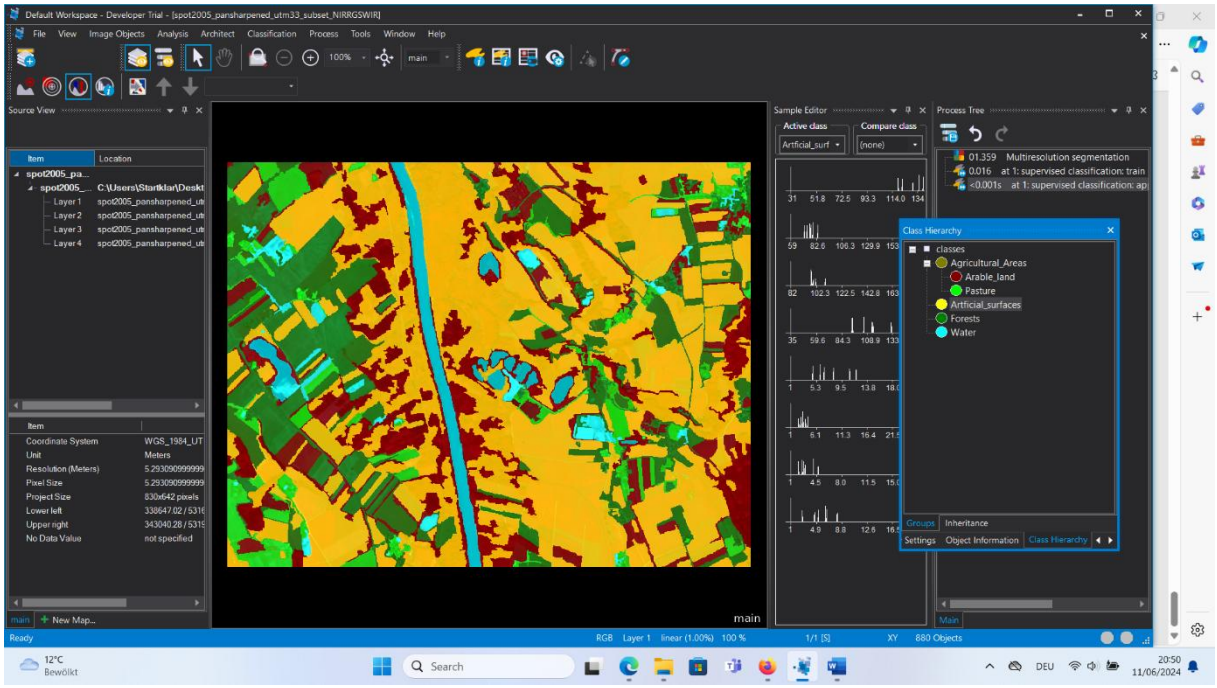


5. What is the meaning of the distance value when you create the feature?

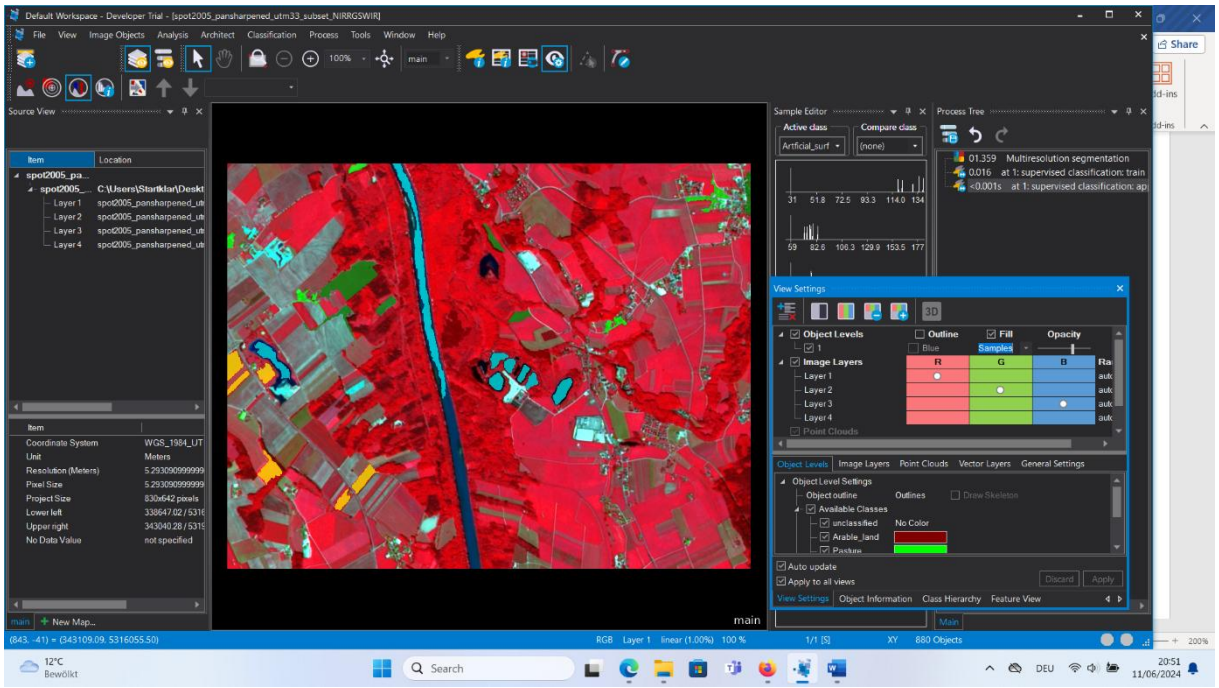
The distance value is calculated by the algorithm [or defined manually] using spectral and spatial characteristics to decide whether the neighbouring objects are to merge or keep separated using their pixel values.

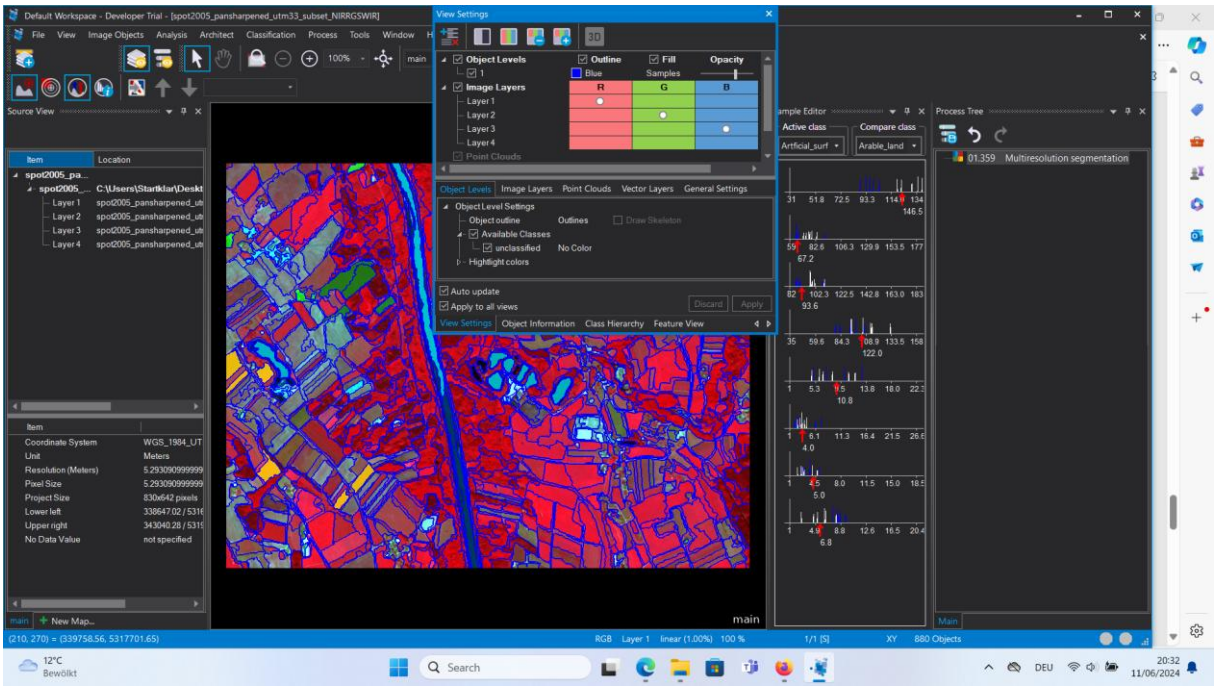
If the distance value between two neighbouring objects is below the defined threshold value, these objects are merged together.

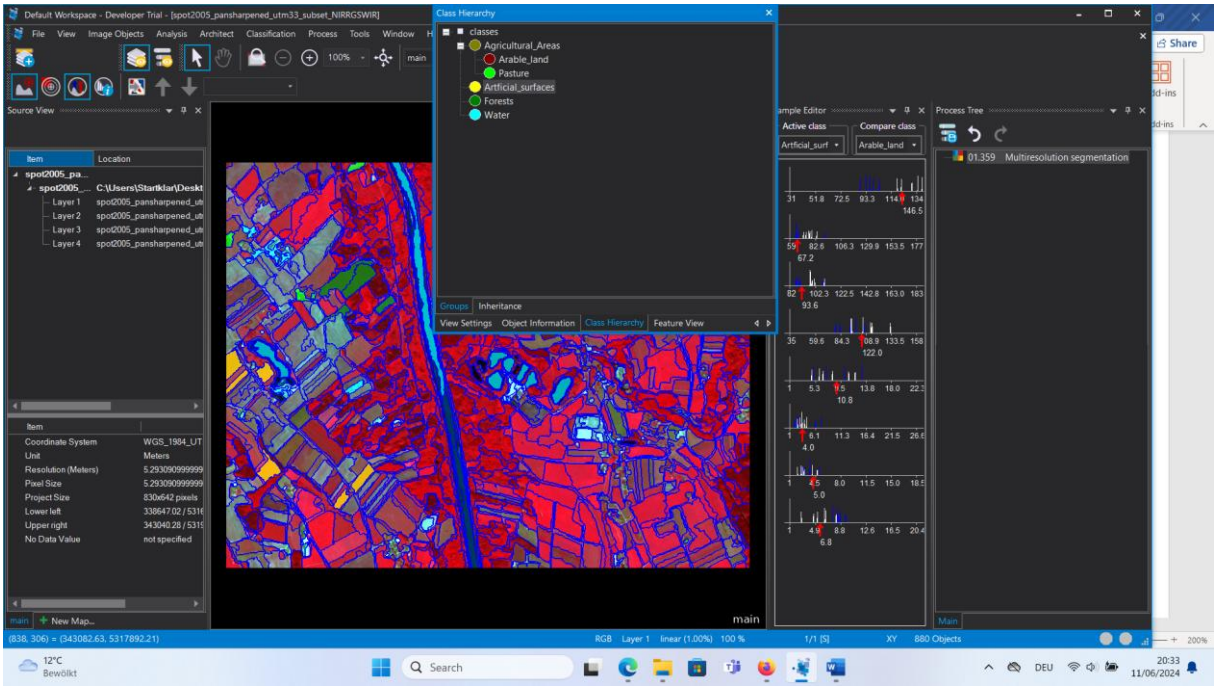
1. Class view-Supervised classification

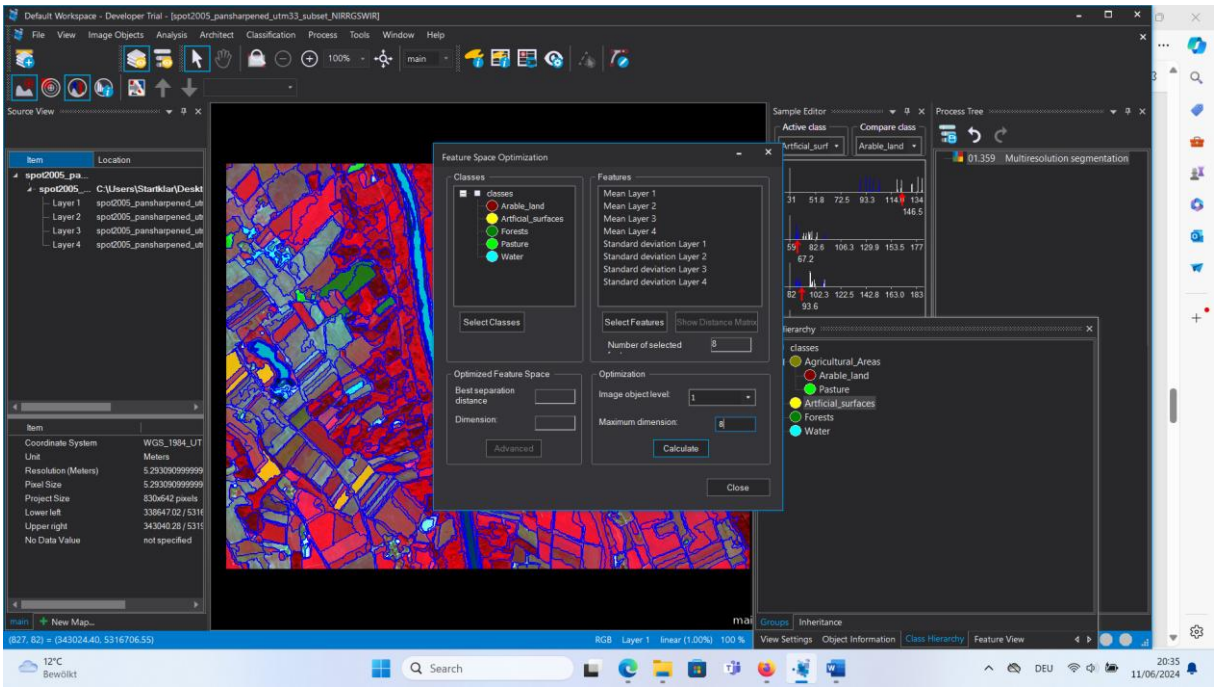


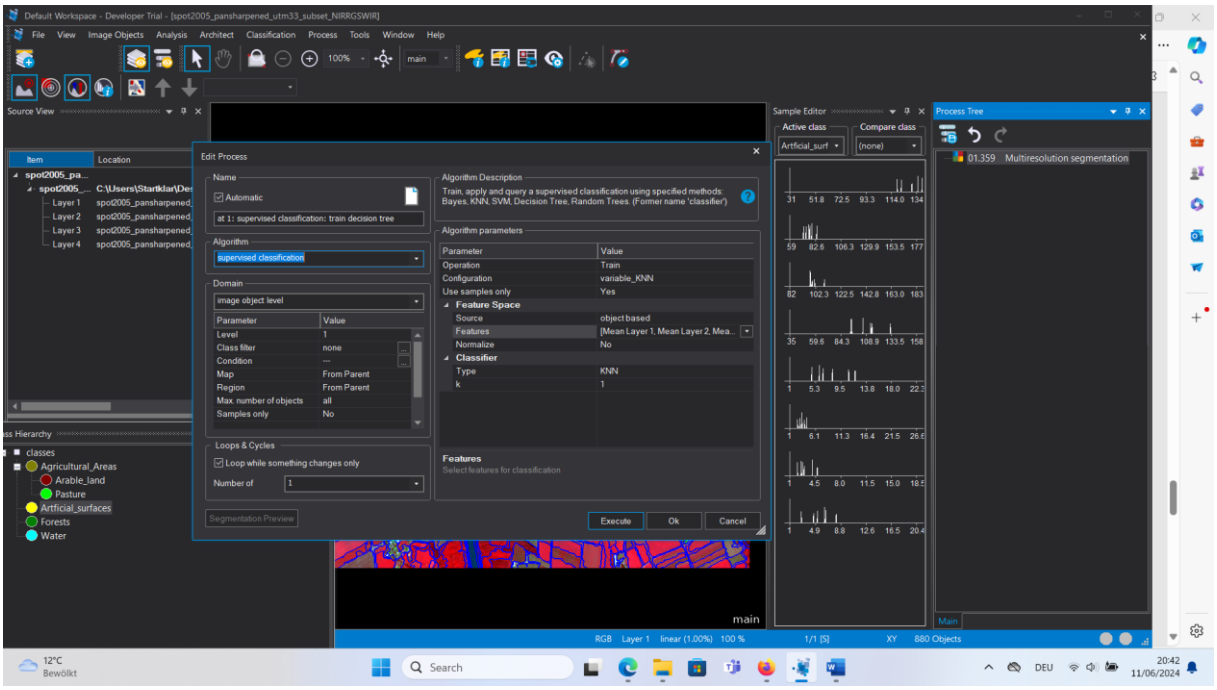
2. Sample view-Supervised classification

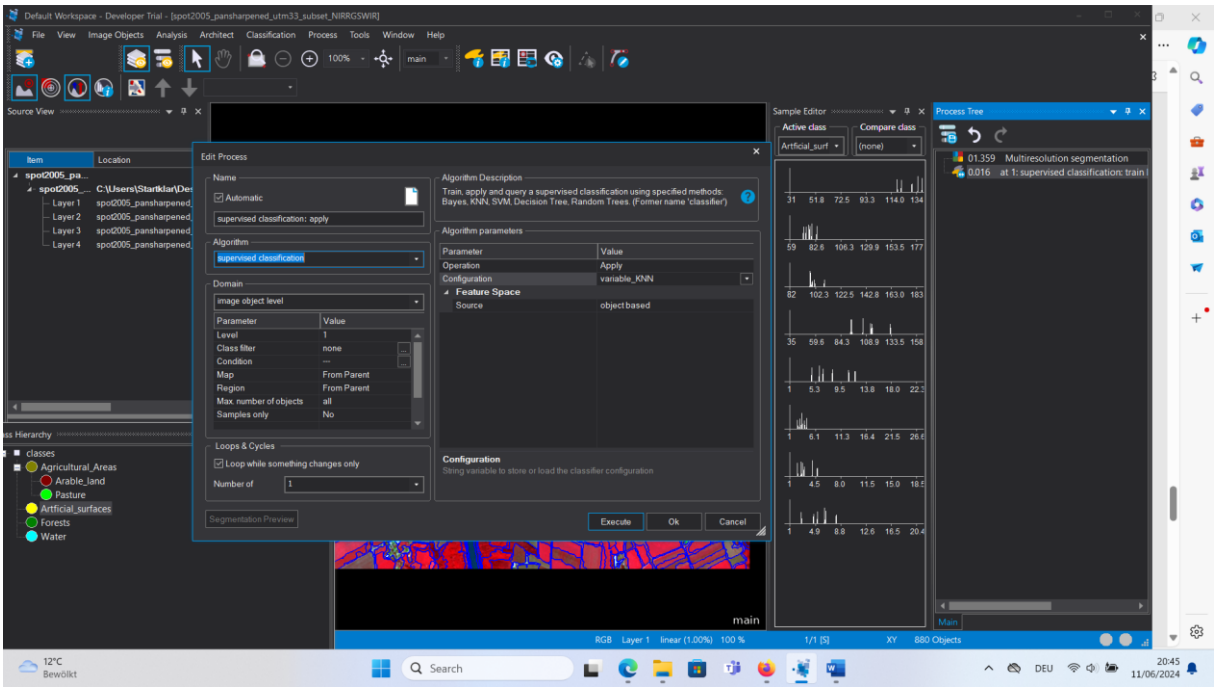


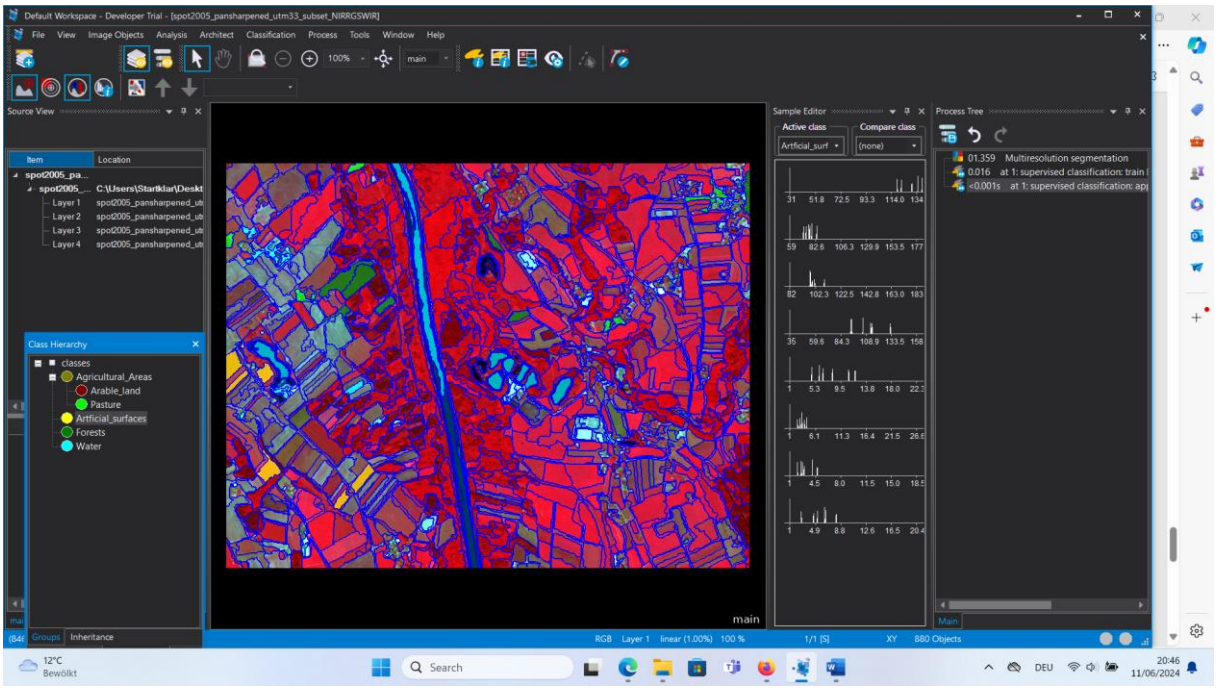

Default Workspace - Developer Trial - [spot2005_pansharpened_utm33_subset_NIRRGSWIR]
View Settings
Object Levels
Image Layers
Layer 1
Layer 2
Layer 3
Layer 4
Point Clouds
Outline
Fill
Opacity
Blue
Samples
R
G
B
Object Level Settings
Object outline
Outlines
Available Classes
unclassified
No Color
Hightlight colors
Auto update
Apply to all views
Discard
Apply
View Settings
Object Information
Class Hierarchy
Feature View
Source View
Item
Location
Coordinate System
WGS_1984_UT
Unit
Meters
Resolution (Meters)
Pixel Size
Project Size
830x642 pixels
Lower left
Upper right
No Data Value
not specified
New Map...
Active class
Compare class
Artficial_surf
Arable_land
Process Tree
01.359 Multiresolution segmentation
main
(210, 270) = (339758.56, 5317701.65)
RGB Layer 1 linear (1.00%) 100 %
1/1 [S]
XY
880 Objects
12°C
Bewölkt
Search
DEU
20:32
11/06/2024

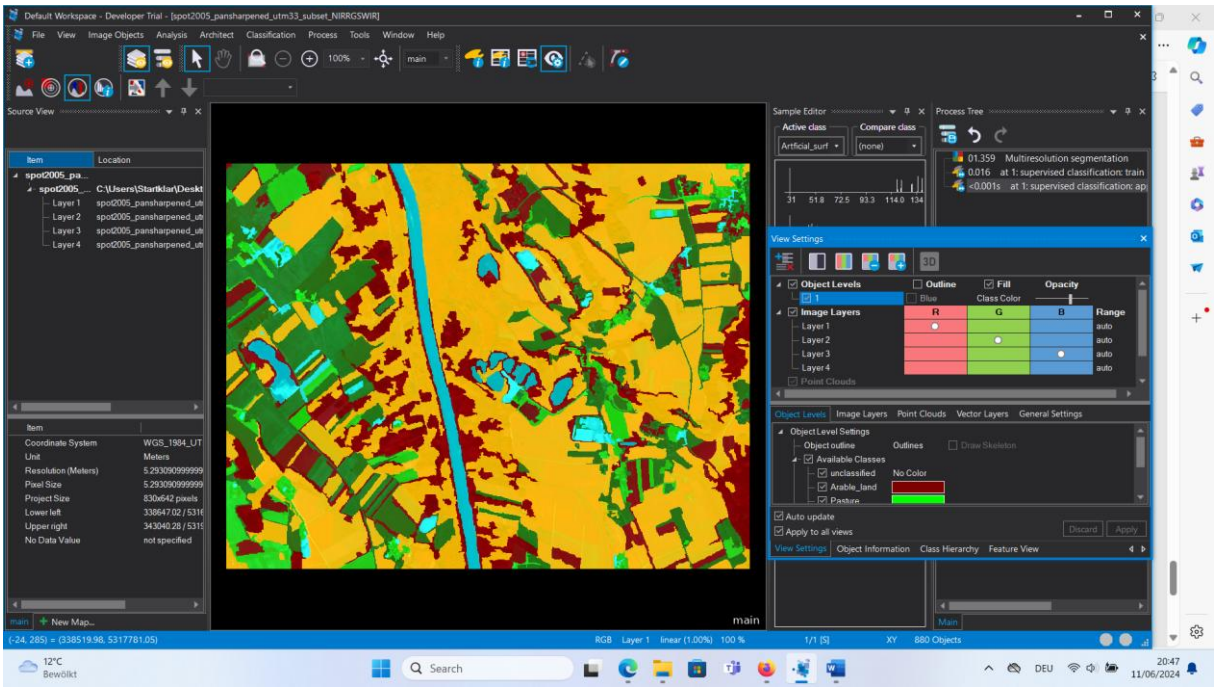

Default Workspace - Developer Trial - [spot2005_pansharpened_utm33_subset_NIRRGSWIR]
Class Hierarchy
classes
Agricultural_Areas
Arable_land
Pasture
Artficial_surfaces
Forests
Water
Groups
Inheritance
View Settings
Object Information
Class Hierarchy
Feature View
Source View
Item
Location
Coordinate System
WGS_1984_UT
Unit
Meters
Resolution (Meters)
Pixel Size
Project Size
830x642 pixels
Lower left
Upper right
No Data Value
not specified
New Map...
Active class
Compare class
Artficial_surf
Arable_land
Process Tree
01.359 Multiresolution segmentation
Share
main
(838, 306) = (343082.63, 5317892.21)
RGB Layer 1 linear (1.00%) 100 %
1/1 [S]
XY
880 Objects
12°C
Bewölkt
Search
DEU
20:33
11/06/2024

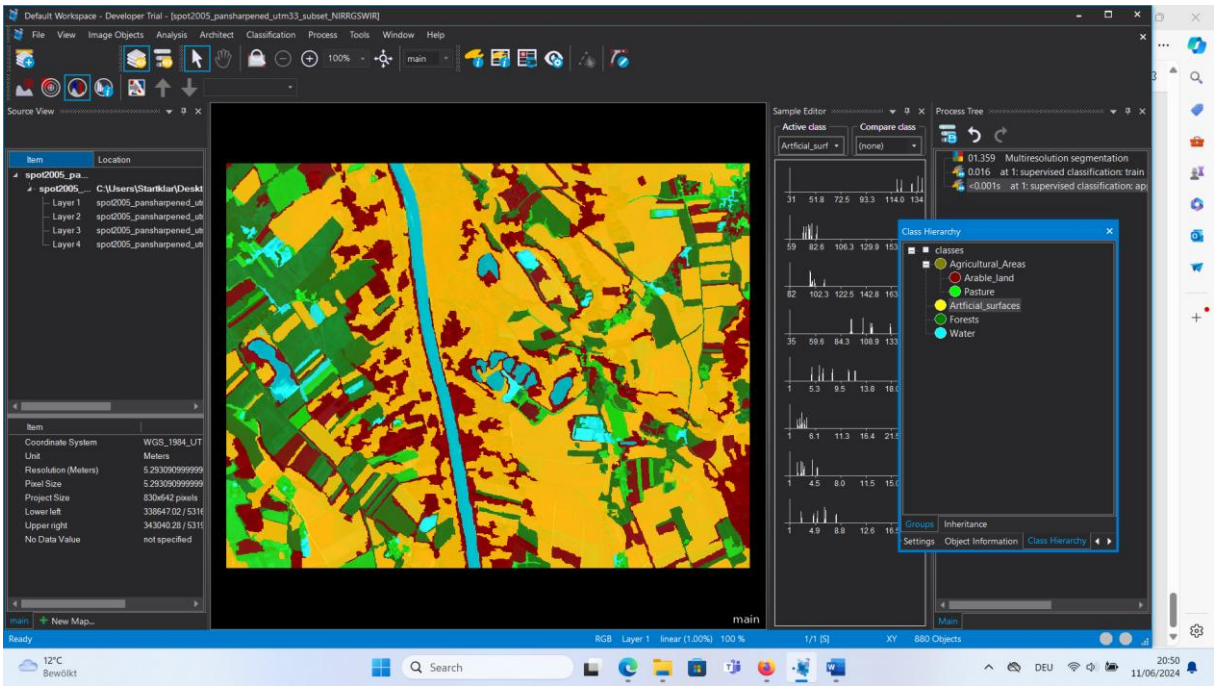

Default Workspace - Developer Trial - [spot2005_pansharpened_utm33_subset_NIRRGSWIR]
File View Image Objects Analysis Architect Classification Process Tools Window Help
Source View
Feature Space Optimization
Classes
classes
Arable_land
Artficial_surfaces
Forests
Pasture
Water
Features
Mean Layer 1
Mean Layer 2
Mean Layer 3
Mean Layer 4
Standard deviation Layer 1
Standard deviation Layer 2
Standard deviation Layer 3
Standard deviation Layer 4
Select Classes
Select Features
Show Distance Matrix
Number of selected
Optimized Feature Space
Best separation distance
Dimension:
Advanced
Optimization
Image object level:
Maximum dimension:
Calculate
Close
Sample Editor
Active class
Compare class
Process Tree
01.359 Multiresolution segmentation
Hierarchy
Agricultural_Areas
Arable_land
Pasture
Artficial_surfaces
Forests
Water
Groups
Inheritance
View Settings
Object Information
Class Hierarchy
Feature View
Search
DEU
20:35
11/06/2024

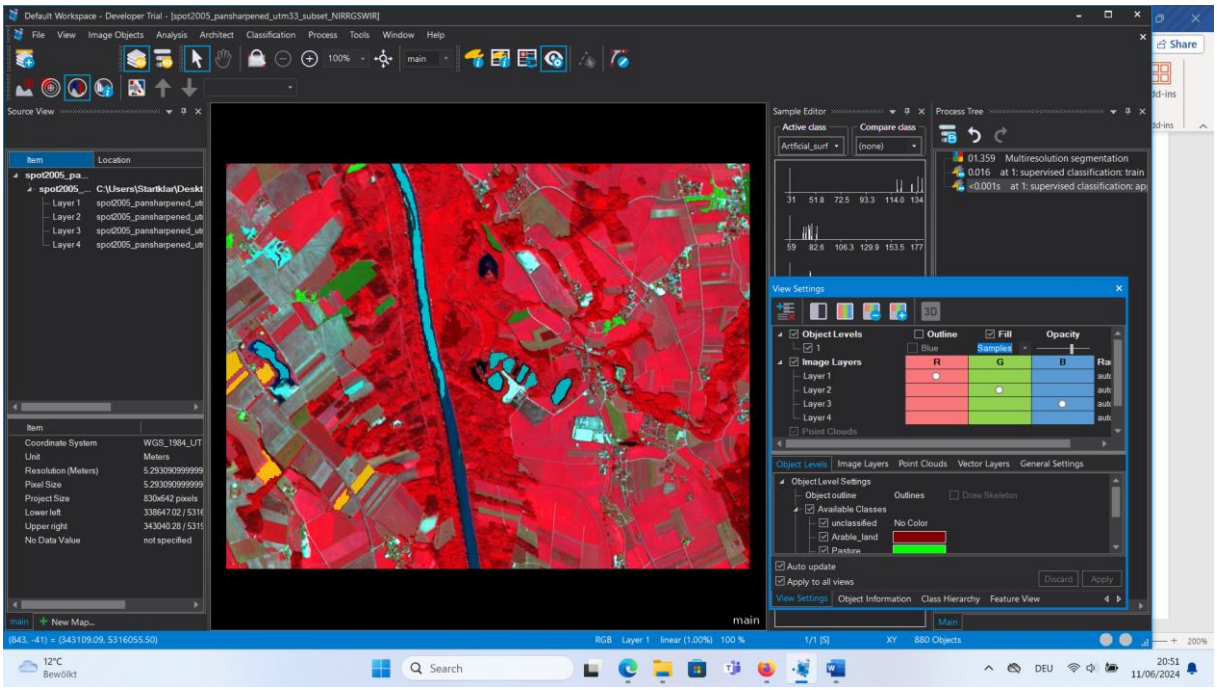

Default Workspace - Developer Trial - [spot2005_pansharpened_utm33_subset_NIRRGSWIR]
File View Image Objects Analysis Architect Classification Process Tools Window Help
Source View
Edit Process
Name
Automatic
Algorithm
supervised classification
Domain
image object level
Parameter
Value
Level
Class filter
Condition
Map
Region
Max. number of objects
Samples only
Loops & Cycles
Loop while something changes only
Number of
Segmentation Preview
Algorithm Description
Train, apply and query a supervised classification using specified methods: Bayes, KNN, SVM, Decision Tree, Random Trees (Former name 'classifier')
Algorithm parameters
Operation
Configuration
Use samples only
Feature Space
Source
Features
Normalize
Classifier
Type
k
Features
Select features for classification
Execute
Ok
Cancel
Sample Editor
Process Tree
01.359 Multiresolution segmentation
Hierarchy
Agricultural_Areas
Arable_land
Pasture
Artficial_surfaces
Forests
Water
Search
DEU
20:42
11/06/2024


Default Workspace - Developer Trial - [spot2005_pansharpened_utm33_subset_NIRRGSWIR]
File View Image Objects Analysis Architect Classification Process Tools Window Help
Source View
Edit Process
Name
Automatic
supervised classification: apply
Algorithm
supervised classification
Domain
image object level
Parameter Value
Level 1
Class filter none
Condition ...
Map From Parent
Region From Parent
Max. number of objects all
Samples only No
Loops & Cycles
Loop while something changes only
Number of 1
Segmentation Preview
Algorithm Description
Train, apply and query a supervised classification using specified methods: Bayes, KNN, SVM, Decision Tree, Random Trees. (Former name 'classifier')
Algorithm parameters
Parameter Value
Operation Apply
Configuration variable_KNN
Feature Space
Source object based
Configuration
String variable to store or load the classifier configuration
Execute Ok Cancel
Class Hierarchy
classes
Agricultural_Areas
Arable_land
Pasture
Artficial_surfaces
Forests
Water
Sample Editor
Active class
Artficial_surf
Compare class
(none)
Process Tree
01.359 Multiresolution segmentation
0.016 at 1: supervised classification: train
main
RGB Layer 1 linear (1.00%) 100 % 1/1 [S] XY 880 Objects
Search
12°C Bewölkt
DEU
20:45 11/06/2024


Default Workspace - Developer Trial - [spot2005_pansharpened_utm33_subset_NIRRGSWIR]
File View Image Objects Analysis Architect Classification Process Tools Window Help
Source View
Class Hierarchy
classes
Agricultural_Areas
Arable_land
Pasture
Artficial_surfaces
Forests
Water
Groups Inheritance
Sample Editor
Active class
Artficial_surf
Compare class
(none)
Process Tree
01.359 Multiresolution segmentation
0.016 at 1: supervised classification: train
<0.001s at 1: supervised classification: ap
main
RGB Layer 1 linear (1.00%) 100 % 1/1 [S] XY 880 Objects
Search
12°C Bewölkt
DEU
20:46 11/06/2024


Default Workspace - Developer Trial - [spot2005_pansharpened_utm33_subset_NIRRGSWIR]
File View Image Objects Analysis Architect Classification Process Tools Window Help
Source View
Sample Editor
Process Tree
View Settings
Object Levels
Image Layers
Point Clouds
Vector Layers
General Settings
Object Level Settings
Available Classes
unclassified
Arable_land
Pasture
Auto update
Apply to all views
Discard
Apply
Object Information
Class Hierarchy
Feature View
main
New Map...
Search
DEU
20:47
11/06/2024


Default Workspace - Developer Trial - [spot2005_pansharpened_utm33_subset_NIRRGSWIR]
File View Image Objects Analysis Architect Classification Process Tools Window Help
Source View
Sample Editor
Process Tree
Class Hierarchy
classes
Agricultural_Areas
Arable_land
Pasture
Artficial_surfaces
Forests
Water
Groups
Inheritance
Settings
Object Information
Class Hierarchy
main
New Map...
Ready
Search
DEU
20:50
11/06/2024


Default Workspace - Developer Trial - [spot2005_pansharpened_utm33_subset_NIRRGSWIR]
File View Image Objects Analysis Architect Classification Process Tools Window Help
Source View
Sample Editor
Process Tree
Active class
Compare class
Artficial_surf
(none)
01.359 Multiresolution segmentation
0.016 at 1: supervised classification: train
<0.001s at 1: supervised classification: ap
View Settings
Object Levels
Outline
Fill
Opacity
Image Layers
Layer 1
Layer 2
Layer 3
Layer 4
Point Clouds
Object Levels
Image Layers
Point Clouds
Vector Layers
General Settings
Object Level Settings
Object outline
Outlines
Draw Skeleton
Available Classes
unclassified
No Color
Arable_land
Auto update
Apply to all views
Discard
Apply
View Settings
Object Information
Class Hierarchy
Feature View
Coordinate System
WGS_1984_UT
Unit
Meters
Resolution (Meters)
Pixel Size
Project Size
830x642 pixels
Lower left
Upper right
No Data Value
not specified
main
New Map...
RGB Layer 1 linear (1.00%) 100 %
1/1 [S]
XY
880 Objects
Share
12°C
Bewölkt
Search
DEU
20:51
11/06/2024